

Day 25 — Logistic Regression Classification

Veda Technology · Data Science Track · Level 2 · Day 25

Objective: Build a Logistic Regression model for binary classification and evaluate predictions on unseen data.

1. Tools

Python · Pandas · NumPy · Scikit-learn · Jupyter Notebook

2. Dataset

Breast Cancer Wisconsin Dataset from scikit-learn. Target classes: 0 = malignant and 1 = benign.

3. Workflow

- Load and inspect data
- Train-test split
- Feature scaling
- Train Logistic Regression
- Predictions and probabilities
- Accuracy and classification report
- Confusion matrix
- Model interpretation

4. Core Code

```
data=load_breast_cancer()
X=pd.DataFrame(data.data,columns=data.feature_names)
y=pd.Series(data.target,name="target")

X_train,X_test,y_train,y_test=train_test_split(
    X,y,test_size=0.20,random_state=42,stratify=y
)

scaler=StandardScaler()
X_train_scaled=scaler.fit_transform(X_train)
X_test_scaled=scaler.transform(X_test)

model=LogisticRegression(max_iter=5000,random_state=42)
model.fit(X_train_scaled,y_train)
y_pred=model.predict(X_test_scaled)
y_proba=model.predict_proba(X_test_scaled)

print(f"Accuracy: {accuracy_score(y_test,y_pred):.4f}")
print(classification_report(y_test,y_pred,target_names=["malignant","benign"]))
```

5. Evaluation

The notebook reports accuracy, precision, recall, F1-score, support, and a confusion matrix.

6. Short Interpretation

Logistic Regression estimates class probabilities. Standardization helps when features have different scales, and `predict_proba()` exposes class probabilities. Preprocessing is fitted only on training data to avoid data leakage.

7. Interview Questions

What is Logistic Regression? A supervised classification algorithm that estimates class probabilities.

Why use it? It is simple, interpretable, probability-based, and useful as a baseline.

What does `predict_proba()` return? Estimated probability for each class.

Why scale features? To put numerical features on comparable scales and support stable optimization.

8. Conclusion

This project demonstrates an end-to-end binary classification workflow: preprocessing, training, prediction, probability estimation, and evaluation.